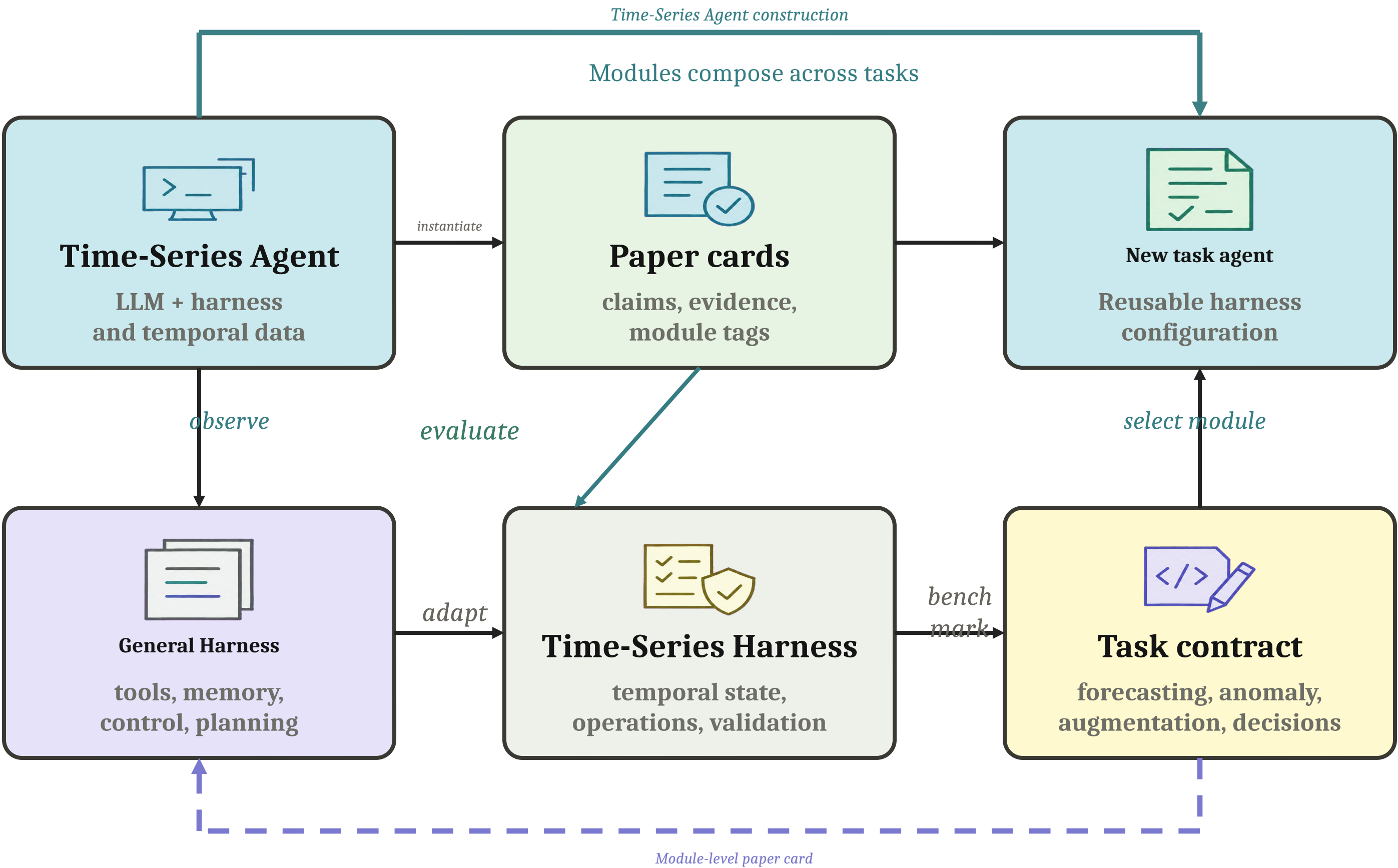


Time-Series Agent system

A module-level construction view



One paper may contribute several modules

Task semantics and benchmark protocols remain explicit

Time-Series Agent

Reader's guide

Survey organization	Sections
§ 1 Introduction	System definition Construction questions
§ 2 What Is a Time-Series Agent?	Review scope Paper-card evidence
§ 3 LLM-Side Contributions	Temporal representation Alignment and reasoning LLM-side methods
§ 4 General Harness	Tools and memory Temporal state Control and execution Provenance and validation
§ 5 General Time-Series Harness	Time-series harnesses Task contracts Reusable modules Benchmark protocols
§ 6 Task-Specific Harness	Forecasting Anomaly diagnosis Augmentation Decision support
§ 7 Infrastructure and Benchmarks	ModernTSF Traces and episodes TimeSeriesGym Reproducible evaluation
§ 8 Open Problems and Case Studies	Paper-card database Open tasks Module-level counts Evidence gaps
§ 9 Conclusion	System map Open problems

Contribution layers in Time-Series Agents

Paper cards record module-level contributions across the LLM, Harness, temporal, task, and evaluation layers.



Research routes in Time-Series Agent systems

Representative module contributions by first public year | LLM and Harness layers

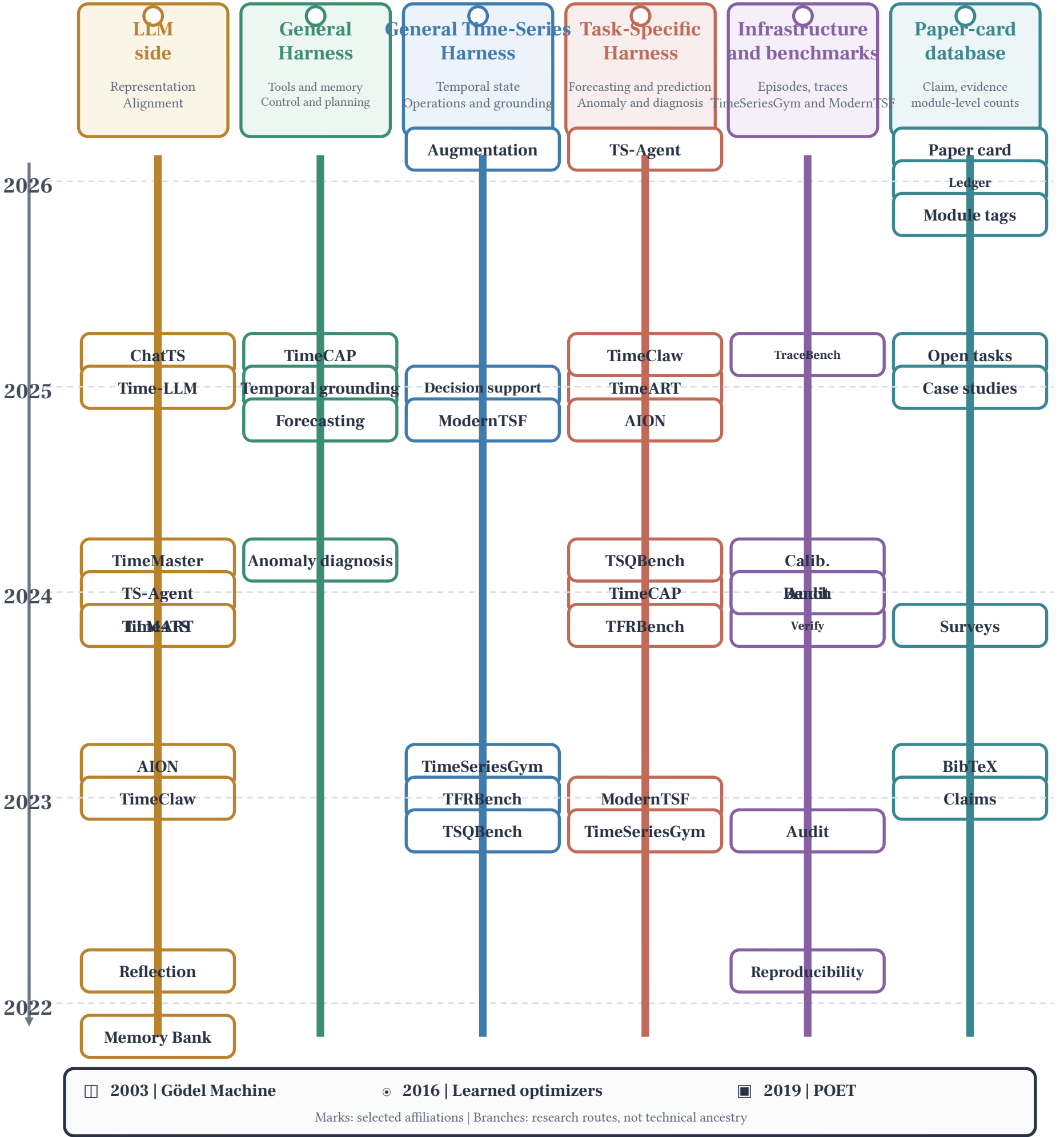


Figure 4. Research routes toward recursive self-improvement. Forty-five representative method papers are grouped into six research routes, with three historical anchors shown separately.

Year bands use the first public dates in the checked source records; spacing within each band is adjusted for legibility. Branch lines denote research families.

Technical ancestry, cross-route transfer, and sustained recursive gains require separate evidence. The selection reflects the literature search through September 12, 2026.

Logos identify one selected author or work-done affiliation per study; “Ind.” denotes independent researchers. Affiliation and logo sources are documented separately.

Forecasting

*origin, horizon,
calibration*

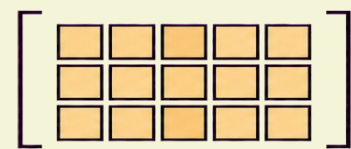


check $\rightarrow$ revise

task-bound

*cross-task
reuse*

same harness



new task

(a) Forecasting

Augmentation

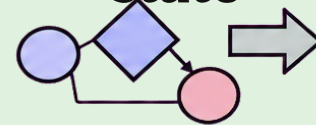
*fidelity, diversity,
utility*



$H_t \rightarrow H_{t+1}$

retain

**Agent + temporal
state**

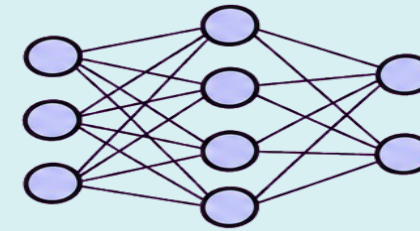


*temporal
memory*

(b) Augmentation

Anomaly diagnosis

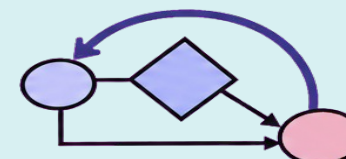
*event, window,
cause*



task output

retain

updated forecast

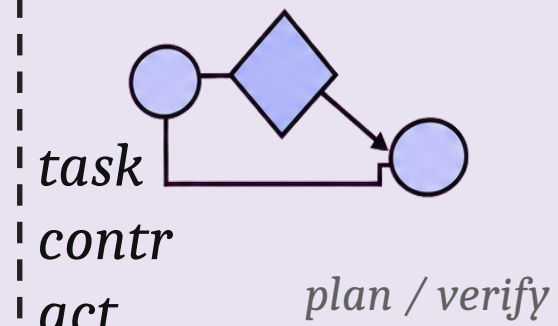


task output

**(c) Anomaly
diagnosis**

Decision support

*action, cost,
uncertainty*



next task episode



new module record

(d) Decision support